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Generalized Composite Probing in Mandarin

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1. Introduction

It is commonly assumed that phrasal movement in natural languages can be strictly classified as either *A-movement* or *Ā-movement*, which are associated with distinct properties: A-movement but not Ā-movement creates new antecedents for anaphor binding and is not subject to weak crossover or Principle C reconstruction; by contrast, Ā-movement but not A-movement is long-distance, crossing c-commanding NPs and finite CPs (see e.g., Postal 1971; Chomsky 1977, 1981; a.o. and Richards 2014 for a comprehensive overview of these properties). From a *featural view of the A/Ā-distinction*, distinct properties associated with A-movement and Ā-movement are derived from distinct [A]-feature (e.g., [ϕ], [CASE], [D]) and [Ā]-feature (e.g., [WH], [REL], [TOP], [FOC]) which trigger A-movement and Ā-movement, respectively (Van Urk 2015; Longenbaugh 2017; Lohninger, Kovač & Wurmbrand 2022; Lohninger & Yip 2023; Chen 2023; a.o.). Furthermore, the possibility of *composite probing* allows for [A]-feature and [Ā]-feature present on the same head to probe together via *conjunctive satisfaction* (Scott 2021), attracting the closest NP with both a matching [A]-feature and a matching [Ā]-feature (Van Urk 2015; Longenbaugh 2017; Lohninger, Kovač & Wurmbrand 2022; Chen 2023; a.o.). The featural view of the A/Ā-distinction and the possibility of composite probing together predict the existence of *composite A/Ā-movement*, triggered by the composite probe [A+Ā], and that such movement should be associated with mixed A/Ā-properties. Positive evidence has been found in Dinka movement to Spec, CP (e.g., topicalization and relativization) (Van Urk 2015), English *tough*-movement (Longenbaugh 2017), and Mandarin passivization (Chen 2023), as summarized in (1). Notably, composite A/Ā-movement is *clause-bound* in English and Mandarin but not in Dinka. This follows from cross-linguistic variations on the distribution of composite probes and a general ban on improper composite A/Ā-movement after Ā-movement. Specifically, it has been proposed that in English and Mandarin, the C head hosts pure Ā-probes but not the composite probe [A+Ā]; consequently, long-distance movement to Spec, CP is necessarily Ā-movement, which cannot be followed by further composite A/Ā-movement (see e.g., Longenbaugh 2017; Chen 2023).[1]

This paper aims to expand cross-linguistic evidence for composite probing, with a specific focus on Mandarin. I argue that composite probing by the composite probe [A+Ā] is generally observed in Mandarin, in the sense that multiple heads projected in the low IP area host the composite probe [A+Ā], and that the Voice head, as a phase head, generally hosts the composite probe [A+Ā] for purposes of successive-cyclic movement. The evidence will come from two types of topicalization and focalization in Mandarin, which exhibit mixed A/Ā-properties, as summarized also in (1). Specifically, I will argue that *IP-internal topicalization and focalization* involve successive-cyclic composite A/Ā-movement via Spec, VoiceP, which terminate at IP-internal Spec, TopP and Spec, FocP, respectively, as illustrated in (2), while *IP-external topicalization and focalization* can involve intermediate steps of composite A/Ā-movement to Spec, VoiceP, followed by a terminating step of Ā-movement to IP-external Spec, TopP and Spec, FocP, respectively, as illustrated in (3).

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[1] The ban on improper composite A/Ā-movement after Ā-movement stems from the ban on improper A-movement after Ā-movement, which in turn stems from the inherent differences between [A]-features and [Ā]-features (see e.g., Neeleman & van De Koot 2010; Obata & Epstein 2011). For more detailed discussion, see Chen (2023).

(1) *Mixed A/Ā-properties of composite A/Ā-movement vs. Mandarin topicalization and focalization*

Properties	A	Ā	Dinka, English, Mandarin composite A/Ā-movement	Mandarin IP-internal topicalization focalization	Mandarin IP-external topicalization focalization
New antecedents for anaphor binding	√	*	√	√	*
No weak crossover	√	*	√	√	* / √
No reconstruction for Principle C	√	*	√	√	* / √
Long-distance ... ... crossing c-commanding NPs	*	√	√	√	√
Long-distance ... ... crossing finite CPs	*	√	√ (D) * (E, M)	*	√

(2) a. *Mandarin: IP-internal topicalization as successive-cyclic composite movement*

$[_{\text{IP}}\ \text{Subj}_{i[\phi]}\ \text{Infl}_{[\phi]}\ [_{\text{TopP}}\ \mathbf{Obj}_{\mathbf{j}[\phi],[\text{TOP}]}\ \text{Top}_{[\phi+\text{TOP}]}\ \cdots\ [_{\text{VoiceP}}\ t_i\ \mathbf{t_j}\ \text{Voice}_{[\phi+\text{TOP}]}\ \cdots\ \mathbf{t_j}\ (...)\ ]]]$

A/Ā-movement A/Ā-movement

b. *Mandarin: IP-internal focalization as successive-cyclic composite movement*

$[_{\text{IP}}\ \text{Subj}_{i[\phi]}\ \text{Infl}_{[\phi]}\ [_{\text{FocP}}\ \mathbf{Obj}_{\mathbf{j}[\phi],[\text{FOC}]}\ \text{Foc}_{[\phi+\text{FOC}]}\ \cdots\ [_{\text{VoiceP}}\ t_i\ \mathbf{t_j}\ \text{Voice}_{[\phi+\text{FOC}]}\ \cdots\ \mathbf{t_j}\ (...)\ ]]]$

A/Ā-movement A/Ā-movement

(3) a. *Mandarin: IP-external topicalization via intermediate composite movement*

$[_{\text{TopP}}\ \mathbf{Obj}_{\mathbf{i}[\phi],[\text{TOP}]}\ \text{Top}_{[\text{TOP}]}\ [_{\text{IP}}\ \text{Subj}_{j[\phi]}\ \text{Infl}_{[\phi]}\ \cdots\ [_{\text{VoiceP}}\ \mathbf{t_i}\ t_j\ \text{Voice}_{[\phi+\text{TOP}]}\ \cdots\ \mathbf{t_i}\ (...)\ ]]]$

Ā-movement A/Ā-movement

b. *Mandarin: IP-external focalization via intermediate composite movement*

$[_{\text{FocP}}\ \mathbf{Obj}_{\mathbf{i}[\phi],[\text{FOC}]}\ \text{Foc}_{[\text{FOC}]}\ [_{\text{IP}}\ \text{Subj}_{j[\phi]}\ \text{Infl}_{[\phi]}\ \cdots\ [_{\text{VoiceP}}\ \mathbf{t_i}\ t_j\ \text{Voice}_{[\phi+\text{FOC}]}\ \cdots\ \mathbf{t_i}\ (...)\ ]]]$

Ā-movement A/Ā-movement

The remaining of the paper is organized as follows: In section 2, I will present the evidence that IP-internal and IP-external topicalization and focalization in Mandarin involve movement into the low IP area and left periphery respectively. In section 3, I will argue that the mixed A/Ā-properties of IP-internal topicalization and focalization are derived from the derivation in (5). In section 4, I will argue that the mixed A/Ā-properties of IP-external topicalization and focalization are derived from the derivation in (3). In section 5, I will account for an additional contrast between IP-internal and IP-external topicalization and focalization in terms of Principle A reconstruction. Section 6 will conclude.

2. Movement into the low IP area and left periphery

In Mandarin, topicalization and focalization can target either an IP-internal or IP-external position, either below or above the grammatical subject in Spec, IP (see e.g., Qu 1994; Shyu 1995; Ting 1995; Paul 2002, 2005; Kuo 2009). In (4-a) and (4-b), the IP-internal or IP-external topic or focus NP is linked to a gap in the post-verbal object position.[2]

(4) a. *IP-internal topicalization/focalization*

$[_{\text{IP}}$ Lisi [(lian) **zhe-jian shi**]$_i$ (dou) diaocha-le $\_\__i$].
Lisi even this-CL matter DIST investigate-PRF

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[2] In the focalization constructions, *lian* 'even' is a focus marker and *dou* is a predicate quantifier, according to Shyu (1995). The syntax and semantics of *dou* have been a topic of much discussion but they are not the focus of this paper; for an overview of the distribution and interpretations of *dou*, see e.g., Xiang (2008).

'Lisi, (even) this matter, investigated (it).'

b. *IP-external topicalization/focalization*
[(Lian) **zhe-jian shi**]$_i$ [$_{IP}$ Lisi (dou) diaocha-le ___$_i$].
even this-CL matter Lisi DIST investigate-PRF
'(Even) this matter, Lisi investigated (it).'

In (5), the IP-internal or IP-external topic or focus NP is linked to an object gap inside a complex NP island; the ill-formedness of (5) suggests that the dependency between the topic or focus NP and the gap is established via movement.

(5) a. *IP-internal topicalization and focalization: Island-sensitive*
*Jingcha [(lian) **Lisi**]$_i$ (dou) zhuazou-le [$_{NP}$ na-ge ___$_j$ da-le ___$_i$ de ren$_j$].
police even Lisi DIST arrest-PRF that-CL hit-PRF REL person
INT: 'The police, (even) Lisi, arrested that person who hit (him).'

b. *IP-external topicalization and focalization: Island-sensitive*
*[(Lian) **Lisi**]$_i$, jingcha (dou) zhuazou-le [$_{NP}$ na-ge ___$_j$ da-le ___$_i$ de ren$_j$].
even Lisi police DIST arrest-PRF that-CL hit-PRF REL person
INT: '(Even) Lisi, the police arrested that person who hit (him).'

Furthermore, in IP-internal topicalization and focalization, the IP-internal topic or focus NP may precede or follow Tense-Aspect-Modality categories and sentential negation, but crucially, an IP-internal topic NP must precede and not follow the progressive aspect *(zheng)zai* or the aspectual/perfective negation *(hai) mei*, while an (IP-internal) focus NP always precedes *dou*, which is outside VoiceP; I take this to indicate that IP-internal topicalization and focalization has variable landing sites within the low IP area, but crucially must be structurally above the VoiceP (cf. Qu 1994; Shyu 1995; Ting 1995; Paul 2002, 2005; Kuo 2009).[3]

(6) *IP-internal topic or focus NP precedes Tense-Aspect-Modality and sentential negation*
a. Lisi [(lian) **zhe-jian shi**]$_i$ zuotian (dou) [$_{VoiceP}$ (haohao) diaocha-le ___$_i$].
Lisi even this-CL matter yesterday DIST well investigate-PRF
'Lisi, (even) this matter, yesterday investigated (it) (well).'

b. Lisi [(lian) **zhe-jian shi**]$_i$ (dou) keneng/keyi [$_{VoiceP}$ (haohao) diaocha ___$_i$].
Lisi even this-CL matter DIST be.possible/be.able well investigate
'Lisi, (even) this matter, is possible/is able to investigate (it) (well).'

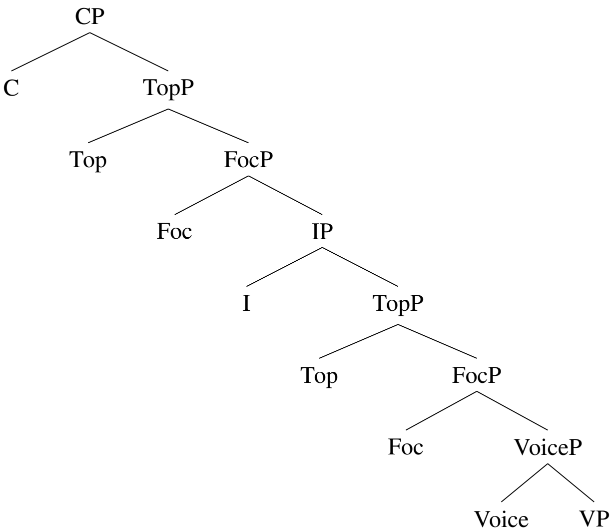
c. Lisi [(lian) **zhe-jian shi**]$_i$ (dou) (zheng)zai/(hai) mei [$_{VoiceP}$ (haohao) diaocha ___$_i$].
Lisi even this-CL matter DIST PROG/yet NEG well investigate
'Lisi, (even) this matter, is investigating/has not investigated (it) (well).'

d. *Lisi (zheng)zai/(hai) mei [**zhe-jian shi**]$_i$ [$_{VoiceP}$ (haohao) diaocha ___$_i$].
Lisi PROG/yet NEG this-CL matter well investigate
INT: 'Lisi, this matter, is investigating/has not investigated (it) (well).'

Hence, I adopt Paul's (2005) proposal that in Mandarin, projections in the low IP area parallel those in the left periphery: IP-internal topicalization and focalization target IP-internal Spec, TopP and Spec, FocP in the low IP area, below the grammatical subject in Spec, IP; IP-external topicalization and focalization target IP-external Spec, TopP and Spec, FocP in the left periphery, above the grammatical subject in Spec, IP, as illustrated in (7).

[3] (IP-internal) topicalization and focalization in Mandarin are subject to the so-called facilitator effects, which requires the movement to cross either a high facilitator outside the VoiceP (e.g., a Tense-Aspect-Modality category) or a low facilitator inside the VoiceP (e.g., a manner adverb) within its path. For more details, see Chen & Yip (2024).

(7) *Mandarin low IP area and left periphery*



3. IP-internal topic and focus movement as successive-cyclic composite movement

It has long been noted that IP-internal topicalization and focalization can establish a long-distance dependency between the IP-internal topic or focus NP and a deeply embedded object gap across non-finite clause boundaries but not across a finite clause boundary (Qu 1994; Ting 1995; Shyu 1995; Kuo 2009):[4]

(8) *IP-internal topicalization and focalization: Restricted long-distance dependency*

a. *Long-distance dependency across non-finite boundaries*

Meiyou-ren, [(lian) **zhe-jian shi**]$_i$, (dou) [$_{VoiceP}$ bi Zhang [$_{VoiceP}$ jiao Wang [$_{VoiceP}$ diaocha __$_i$]]].
no-person even this-CL matter DIST force Zhang ask Wang investigate
'Nobody, (even) this matter$_i$, forced Zhang to ask Wang to investigate (it$_i$).'

b. *No long-distance dependency across finite boundary*

*Meiyou-ren, [(lian) **zhe-jian shi**]$_i$, (dou) xiangxin/zhidao [$_{CP}$ Zhang hui diaocha __$_i$].
no-person even this-CL matter DIST believe/know Zhang will investigate
INT: 'Nobody, (even) this matter$_i$, believes/knows that Zhang will investigate (it$_i$).'

Furthermore, (long-distance) IP-internal topicalization and focalization exhibit properties of A-movement, creating new antecedents for anaphor binding and not subject to weak crossover or Principle C reconstruction (Qu 1994; Ting 1995; Shyu 1995; Kuo 2009).

(9) a. *IP-internal topicalization and focalization: New antecedents for anaphor binding*

Zhang, [(lian) **zhe-ge ren**]$_i$, (dou) [$_{VoiceP}$ bi ta-ziji$_i$-de pengyou [$_{VoiceP}$ ma-guo __$_i$]].
Zhang even that-CL person DIST force 3SG-self's friend scold-EXP
'Zhang, (even) this person$_i$, once forced his$_i$ friend to scold (him$_i$).'

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[4] In (8), *meiyou-ren* 'nobody' resists topicalization; hence, (8-b) cannot have a derivation where (i) the embedded object undergoes IP-external topicalization to precede the matrix subject, and (ii) the matrix subject undergoes further topicalization to precede the topicalized embedded object.

b. *IP-internal topicalization (of quantifier phrase): No weak crossover*

Zhang, [**mei-ge ren**]$_{i}$, dou $[_{\text{VoiceP}}$ bi ta$_{i}$-de pengyou $[_{\text{VoiceP}}$ ma-guo __$_{i}$]]?
Zhang every-CL person DIST force 3SG's friend scold-EXP
Lit. 'Zhang, every person$_{i}$, once forced his$_{i}$ friend to scold (him$_{i}$)?'

c. *IP-internal focalization: No weak crossover*

Zhang, [lian **Li**]$_{i}$, dou $[_{\text{VoiceP}}$ bi ta$_{i}$-de pengyou $[_{\text{VoiceP}}$ ma-guo __$_{i}$]]?
Zhang even Li DIST force 3SG's friend scold-EXP
Lit. 'Zhang, even Li$_{i}$, once forced his$_{i}$ friend to scold (him$_{i}$).'

d. *IP-internal topicalization and focalization: No Principle C reconstruction*

Zhang, [(lian) **Li$_{i}$-de pengyou**]$_{j}$, (dou) $[_{\text{VoiceP}}$ bi ta$_{i}$ $[_{\text{VoiceP}}$ ma-guo __$_{j}$]].
Zhang even Li's friend DIST force 3SG scold-EXP
'Zhang, (even) Li$_{i}$'s friend$_{j}$, once forced him$_{i}$ to scold (him$_{j}$).'

In contrast to existing work that analyzes IP-internal topicalization and focalization as instances of A-movement, thereby overlooking their lack of A-minimality/locality effects and the presence of information-structural effects, I propose that IP-internal topicalization and focalization involve successive-cyclic composite A/Ā-movement via Spec, VoiceP, terminating at IP-internal Spec, TopP, and Spec, FocP, respectively, as illustrated in (10). This derivation necessitates that both the Voice head, as a phase head, and the IP-internal Top and Foc heads host the composite probes [ϕ + TOP] and [ϕ + FOC], and it follows that this composite A/Ā-movement should exhibit mixed A/Ā-properties.

(10) a. *Mandarin: IP-internal topicalization as successive-cyclic composite movement*

$[_{\text{IP}}$ Subj$_{i[\phi]}$ Infl$_{[\phi]}$ $[_{\text{TopP}}$ **Obj**$_{\mathbf{j}[\phi],[\text{TOP}]}$ Top$_{[\phi+\text{TOP}]}$ ... $[_{\text{VoiceP}}$ t$_{i}$ **t$_{j}$** Voice$_{[\phi+\text{TOP}]}$ ... **t$_{j}$** (...)]]]
A/Ā-movement A/Ā-movement

b. *Mandarin: IP-internal focalization as successive-cyclic composite movement*

$[_{\text{IP}}$ Subj$_{i[\phi]}$ Infl$_{[\phi]}$ $[_{\text{FocP}}$ **Obj**$_{\mathbf{j}[\phi],[\text{FOC}]}$ Foc$_{[\phi+\text{FOC}]}$ ... $[_{\text{VoiceP}}$ t$_{i}$ **t$_{j}$** Voice$_{[\phi+\text{FOC}]}$ ... **t$_{j}$** (...)]]]
A/Ā-movement A/Ā-movement

Furthermore, it is important that the Mandarin C head, also as a phase head, only hosts *pure Ā-probes* (e.g., [TOP], [FOC]) and not the composite probes [ϕ + TOP] and [ϕ + FOC].[5] Consequently, Ā-movement to Spec, CP cannot feed composite A/Ā-movement to the IP-internal Spec, TopP and Spec, FocP, due to the ban on improper composite A/Ā-movement after Ā-movement (see e.g., Longenbaugh 2017; Chen 2023).

(11) *Improper composite A/Ā-movement after Ā-movement*

a. *$[_{\text{IP}}$ Subj$_{i[\phi]}$ Infl$_{[\phi]}$ $[_{\text{TopP}}$ **Obj**$_{\mathbf{j}[\phi],[\text{TOP}]}$ Top$_{[\phi+\text{TOP}]}$ ... $[_{\text{VoiceP}}$ t$_{i}$ **t$_{j}$** ... $[_{\text{CP}}$ **t$_{j}$** C$_{[\text{TOP}]}$ ... **t$_{j}$** (...)]]]]
A/Ā-movement ... Ā-movement

b. *$[_{\text{IP}}$ Subj$_{i[\phi]}$ Infl$_{[\phi]}$ $[_{\text{FocP}}$ **Obj**$_{\mathbf{j}[\phi],[\text{FOC}]}$ Foc$_{[\phi+\text{FOC}]}$ ... $[_{\text{VoiceP}}$ t$_{i}$ **t$_{j}$** ... $[_{\text{CP}}$ **t$_{j}$** C$_{[\text{FOC}]}$ ... **t$_{j}$** (...)]]]]
A/Ā-movement ... Ā-movement

4. IP-external topic and focus movement via intermediate composite movement

Unlike IP-internal topicalization and focalization, which are clause-bound, and like typical Ā-movement, IP-external topicalization and focalization can establish a long-distance dependency between the IP-external topic or focus NP and a deeply embedded object gap across a finite clause boundary:

[5] In Chen (2023), it is argued that the Mandarin C head also hosts a *pure ϕ-probe*, which is active in the case of hyperraising to subject via Spec, CP. Hence, both an [A]-feature and [Ā]-features are present on the Mandarin C head, but, crucially, they probe separately and independently.

(12) *IP-external topicalization and focalization: Long-distance dependency across finite boundary*
[(Lian) **zhe-jian shi**]$_i$, Li (dou) xiangxin/zhidao [$_{CP}$ Zhang hui diaocha __$_i$].
even this-CL matter Li DIST believe/know Zhang will investigate
'(Even) this matter$_i$, Li believes/knows that Zhang will investigate (it$_i$).'

Also unlike IP-internal topicalization and focalization, IP-external topicalization and focalization do not create new antecedents for anaphor binding, regardless of whether the anaphor is present between the IP-external topic or focus NP and the matrix VoiceP, as in (13-a), or between the embedded CP and the embedded VoiceP, as in (13-b), or between the matrix VoiceP and the embedded VoiceP, as in (13-c) (cf. Qu 1994; Ting 1995; Shyu 1995; Kuo 2009).

(13) *IP-external topicalization and focalization: No new antecedents for anaphor binding*

a. *... anaphor **between topic/focus and matrix VoiceP***
*[(Lian) **Li**]$_i$, ta-ziji$_i$-de pengyou (dou) [$_{VoiceP}$ ma-guo __$_i$].
even Li 3SG-self's friend DIST scold-EXP
INT: '(Even) Li$_i$, his$_i$ friend once scolded (him$_i$).'

b. *... anaphor **between embedded CP and embedded VoiceP***
*[(Lian) **Li**]$_i$, Zhang (dou) xiangxin/zhidao [$_{CP}$ ta-ziji$_i$-de pengyou [$_{VoiceP}$ ma-guo __$_i$]].
even Li Zhang DIST believe/know 3SG-self's friend scold-EXP
INT: '(Even) Li$_i$, Zhang believes/knows that his$_i$ friend once scolded (him$_i$).'

c. *... anaphor **between matrix VoiceP and embedded VoiceP***
*[(Lian) **Li**]$_i$, Zhang (dou) [$_{VoiceP}$ bi ta-ziji$_i$-de pengyou [$_{VoiceP}$ ma-guo __$_i$]].
even Li Zhang DIST force 3SG-self's friend scold-EXP
INT: '(Even) Li$_i$, Zhang once forced his$_i$ friend to scold (him$_i$).'

More interestingly, whether IP-external topicalization and focalization are subject to weak crossover and Principle C reconstruction depends on the position of the pronoun co-indexed with the IP-external topic or focus NP in the given configurations (cf. Qu 1994).[6] Specifically, IP-external topicalization and focalization are subject to weak crossover, when the co-indexed pronoun is outside the VoiceP phase(s), as in (14-a), (14-b) and (15-a), (15-b) (cf. Qu 1994; Ting 1995; Shyu 1995; Kuo 2009), but is immune to weak crossover, when the co-indexed pronoun is inside the VoiceP phase(s), as in (14-c) and (15-c) (cf. Qu 1994).

(14) *IP-external topicalization (of quantifier phrase): (No) weak crossover*

a. *... co-indexed pronoun **between topic and matrix VoiceP***
*[**Mei-ge ren**]$_i$, ta$_i$-de pengyou (dou) [$_{VoiceP}$ ma-guo __$_i$]].
every-CL person 3SG's friend DIST scold-EXP
INT: 'Every person$_i$, his$_i$ friend once scolded (him$_i$).'

b. *... co-indexed pronoun **between embedded CP and embedded VoiceP***
*[**Mei-ge ren**]$_i$, Zhang (dou) xiangxin/zhidao [$_{CP}$ ta$_i$-de pengyou [$_{VoiceP}$ ma-guo __$_i$]].
every-CL person Zhang DIST believe/know 3SG's friend scold-EXP
INT: 'Every person$_i$, Zhang believes/knows that his$_i$ friend once scolded (him$_i$).'

c. *... co-indexed pronoun **between matrix VoiceP and embedded VoiceP***
[**Mei-ge ren**]$_i$, Zhang (dou) [$_{VoiceP}$ bi ta$_i$-de pengyou [$_{VoiceP}$ ma-guo __$_i$]].
every-CL person Zhang DIST force 3SG's friend scold-EXP

[6] Qu (1994) fails to identify the link between the presence or absence of weak crossover and Principle C reconstruction and the position of the co-indexed pronoun and simply concludes that IP-external topicalization and focalization may be instances of either A-movement, when they are immune to weak crossover and Principle C reconstruction, or Ā-movement, when they are subject to weak crossover and Principle C reconstruction. This is a logical flaw: if IP-external topicalization and focalization can be instances of A-movement, then any ungrammaticality due to weak crossover or Principle C reconstruction should be circumventable, contrary to fact.

Lit. 'Every person$_i$, Zhang once forced his$_i$ friend to scold (him$_i$).'

(15) *IP-external focalization: (No) weak crossover*

a. *... co-indexed pronoun* ***between focus and matrix VoiceP***
*[Lian **Li**]$_i$, ta$_i$-de pengyou (dou) [$_{\text{VoiceP}}$ ma-guo __$_i$]].
even Li 3SG's friend DIST scold-EXP
INT: 'Even Li$_i$, his$_i$ friend once scolded (him$_i$).'

b. *... co-indexed pronoun* ***between embedded CP and embedded VoiceP***
*[Lian **Li**]$_i$, Zhang (dou) xiangxin/zhidao [$_{\text{CP}}$ ta$_i$-de pengyou [$_{\text{VoiceP}}$ ma-guo __$_i$]].
even Li Zhang DIST believe/know 3SG's friend scold-EXP
INT: 'Even Li$_i$, Zhang believes/knows that his$_i$ friend once scolded (him$_i$).'

c. *... co-indexed pronoun* ***between matrix VoiceP and embedded VoiceP***
[Lian **Li**]$_i$, Zhang (dou) [$_{\text{VoiceP}}$ bi ta$_i$-de pengyou [$_{\text{VoiceP}}$ ma-guo __$_i$]].
even Li Zhang DIST force 3SG's friend scold-EXP
Lit. 'Even Li$_i$, Zhang once forced his$_i$ friend to scold (him$_i$).'

Similarly, IP-external topicalization and focalization show (weak) Principle C reconstruction effects, when the co-indexed pronoun is outside the VoiceP phase(s), as in (16-a) and (16-b),[7] but show no Principle C reconstruction effects, when the co-indexed pronoun is inside the VoiceP phase(s), as in (16-c) (cf. Qu 1994).

(16) *IP-external topicalization and focalization: (No) Principle C reconstruction*[8]

a. *... co-indexed pronoun* ***between topic/focus and matrix VoiceP***
*/?? [(Lian) **zhe-ge ren$_i$-de pengyou**]$_j$, ta$_i$ (dou) [$_{\text{VoiceP}}$ ma-guo __$_j$]].
even this-CL person's friend 3SG DIST scold-EXP
INT: '(Even) this person$_i$'s friend$_j$, he$_i$ once scolded (him$_j$).'

b. *... co-indexed pronoun* ***between embedded CP and embedded VoiceP***
*/? [(Lian) **zhe-ge ren$_i$-de pengyou**]$_j$, Zhang (dou) xiangxin/zhidao [$_{\text{CP}}$ ta$_i$ [$_{\text{VoiceP}}$
even this-CL person's friend Zhang DIST believe/know 3SG
ma-guo __$_j$]].
scold-EXP
'(Even) this person$_i$'s friend$_j$, Zhang believes/knows that he$_i$ once scolded (him$_j$).'

c. *... co-indexed pronoun* ***between matrix VoiceP and embedded VoiceP***
[(Lian) **zhe-ge ren$_i$-de pengyou**]$_j$, Zhang (dou) [$_{\text{VoiceP}}$ bi ta$_i$ [$_{\text{VoiceP}}$ ma-guo __$_j$]].
even this-CL person's friend Zhang DIST force 3SG scold-EXP
'(Even) this person$_i$'s friend$_j$, Zhang once forced him$_i$ to scold (him$_j$).'

I propose that IP-external topicalization and focalization in Mandarin, which otherwise involve typical successive-cyclic Ā-movement via VoiceP and CP phase edges, can also involve intermediate steps of composite A/Ā-movement to Spec, VoiceP, followed by a terminating step of Ā-movement to IP-external Spec, TopP and Spec, FocP. The intermediate composite A/Ā-movement to Spec, VoiceP is associated with mixed A/Ā-properties (e.g., the lack of weak crossover/Principle C reconstruction), while the terminating Ā-movement to IP-external Spec, TopP and Spec, FocP is associated with Ā-properties (e.g., weak crossover/Principle C reconstruction).

[7] In (16-a), Principle C reconstruction effect is weak with topicalization and stronger with focalization. In (16-b), Principle C reconstruction effect is further weakened with topicalization (cf. Huang 1993: ex. 54c), which might indicate that cross-clausal reconstruction for Principle C is optional with topicalization.

(17) a. *Mandarin: IP-external topicalization via intermediate composite movement*

$[_{\text{TopP}}\ \mathbf{Obj}_{\mathbf{i}[\phi],[\textsc{top}]}\ \text{Top}_{[\textsc{top}]}\ [_{\text{IP}}\ \text{Subj}_{j[\phi]}\ \text{Infl}_{[\phi]}\ \dots\ [_{\text{VoiceP}}\ \mathbf{t_i}\ t_j\ \text{Voice}_{[\phi+\textsc{top}]}\ \dots\ \mathbf{t_i}\ (\dots)\ ]]]$

Ā-movement A/Ā-movement

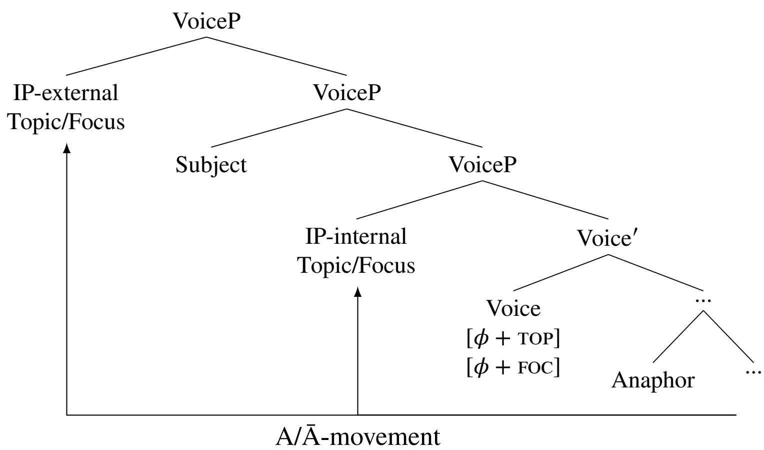
b. *Mandarin: IP-external focalization via intermediate composite movement*

$[_{\text{FocP}}\ \mathbf{Obj}_{\mathbf{i}[\phi],[\textsc{foc}]}\ \text{Foc}_{[\textsc{foc}]}\ [_{\text{IP}}\ \text{Subj}_{j[\phi]}\ \text{Infl}_{[\phi]}\ \dots\ [_{\text{VoiceP}}\ \mathbf{t_i}\ t_j\ \text{Voice}_{[\phi+\textsc{foc}]}\ \dots\ \mathbf{t_i}\ (\dots)\ ]]]$

Ā-movement A/Ā-movement

Recall that IP-internal topicalization and focalization can, while IP-external topicalization and focalization cannot, create new antecedents for anaphor binding, even when the anaphor is inside a VoiceP phase. I propose that at the edge of a VoiceP phase, the external argument is the SUBJECT that delimits the binding domain for anaphors. Furthermore, IP-internal topicalization and focalization stop off at an *inner* Spec, VoiceP, below the thematic position of the external argument, while IP-external topicalization and focalization stop off at an *outer* Spec, VoiceP, above the thematic position of the external argument; in this way, the linear orderings of phrases established by phase-by-phase spell-out are consistent throughout a derivation (see e.g., Fox & Pesetsky 2005; Davis 2020). Consequently, in the former case, the topic or focus NP is *inside* the binding domain for anaphors that lack a more locally accessible SUBJECT, while in the latter case, the topic or focus NP is *outside* the binding domain for anaphors that are bound by the external argument in Spec, VoiceP/its thematic position, the more locally accessible SUBJECT.

(18) *Topicalization/Focalization and anaphor binding at VoiceP phase*



5. A note on Principle A reconstruction

In Mandarin, there is also a contrast between IP-internal and IP-external topicalization and focalization in terms of Principle A reconstruction effects (see e.g., Qu 1994; Ting 1995; Shyu 1995; Kuo 2009):

(19) a. *IP-internal topicalization and focalization: no Principle A reconstruction*

Zhangsan$_i$ [(lian) **ta-ziji$_{i/*j}$-de pengyou**] (dou) bi Lisi$_j$ ma-guo __.
Zhangsan even 3SG-self's friend DIST force Lisi scold-EXP
'Zhangsan$_i$, (even) his$_{i/*j}$ friend, forced Lisi$_j$ to scold (him).'

b. *IP-external topicalization and focalization: Principle A reconstruction*

[(Lian) **ta-ziji$_{i/j}$-de pengyou**], Zhangsan$_i$ (dou) bi Lisi$_j$ ma-guo __.
even 3SG-self's friend Zhangsan DIST force Lisi scold-EXP
'(Even) his$_{i/j}$ friend, Zhangsan$_i$ forced Lisi$_j$ to scold (him).'

Cross-linguistically, there is variation in whether A-movement shows Principle A reconstruction effects. In English, both A-movement and Ā-movement show Principle A reconstruction effects (Belletti & Rizzi 1988; Pesetsky 2013), while in Dutch, only Ā-movement shows Principle A reconstruction effects (see e.g., Neeleman & Van De Koot 2010).

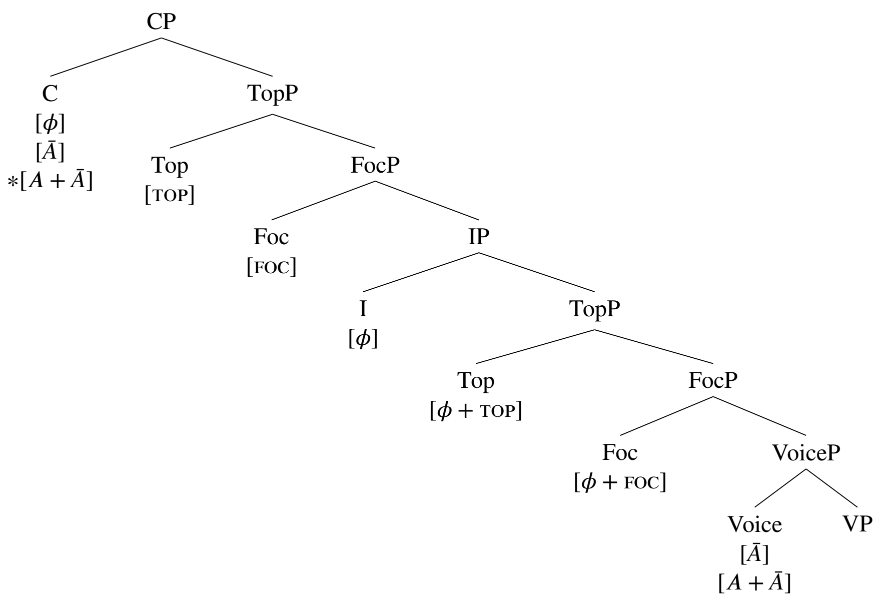
I propose that Mandarin is unlike English and like Dutch in that only Ā-movement shows Principle A reconstruction effects. The contrast in (19) follows because IP-internal topicalization and focalization involve successive-cyclic composite A/Ā-movement, which does not show Principle A reconstruction effects, while IP-external topicalization and focalization can involve typical successive-cyclic Ā-movement, which shows Principle A reconstruction effects.[9]

6. Conclusion

To recapitulate, this paper has aimed to expand cross-linguistic evidence for composite probing, with a specific focus on Mandarin. I have argued that composite probing by the composite probe [A+Ā] is generally observed in Mandarin, in the sense that multiple heads projected in the low IP area host the composite probe [A+Ā], and that the Voice head, as a phase head, generally hosts the composite probe [A+Ā] for purposes of successive-cyclic movement. The evidence has come from two types of topicalization and focalization in Mandarin, which exhibit mixed A/Ā-properties. Specifically, I have argued that IP-internal topicalization and focalization involve successive-cyclic composite A/Ā-movement via Spec, VoiceP, which terminate at IP-internal Spec, TopP and Spec, FocP, respectively, while IP-external topicalization and focalization can involve an intermediate step of composite A/Ā-movement to Spec, VoiceP, followed by a terminating step of Ā-movement to IP-external Spec, TopP and Spec, FocP, respectively.

The proposed analyses lead to the distribution of A-, Ā-, and composite probes in Mandarin low IP area and left periphery in (20). The IP-internal Top head and Foc head host composite A/Ā-probes, [ϕ+TOP] and [ϕ+FOC] respectively, which trigger IP-internal topicalziation and focalization, while the IP-external Top head and Foc head host pure Ā-probes, [TOP] and [FOC] respectively, which trigger IP-external topicalziation and focalization. The Voice head, as a phase head, hosts both composite A/Ā-probes and pure Ā-probes; by contrast, the C head, also as a phase head, does not host composite A/Ā-probes.

(20) *Distribution of A-, Ā-, and composite probes in Mandarin low IP area and left periphery*



[9] Given the above proposal that the external argument in Spec, VoiceP/its thematic position can bind an anaphor before its A-movement to Spec, IP, it remains puzzling as to why an anaphor cannot be bound at its thematic position before (A-movement and) composite A/Ā-movement. I leave this puzzle and an account of the cross-linguistic variation on Principle A reconstruction effects to future research.

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